ABSTRACT

How cultural differences impacts information-seeking behavior and the recollection of information on websites

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Significant differences were observed in the type of cognitive processing employed by individuals from different cultures. Western cultures, known for their individualistic tendencies, tend to adopt analytic processing, focusing on specific elements while overlooking broader context. On the other hand, East Asian cultures, often characterized as more collectivist, demonstrate holistic processing, emphasizing the entire visual scene and its contextual relevance. These insights have been applied in website design research, where it was found that Japanese participants, in comparison to their US counterparts, took significantly longer to locate information on websites. To expand on these findings, our study encompassed a diverse range of countries from both East Asia and the West. We did not identify a significant effect between individuals from Eastern countries and Western countries on how they search information on a webpage, particularly when targeting foreground and background areas. We discussed how designers can enhance user experiences and optimize the

information retrieval process by tailoring websites to accommodate diverse cultural preferences.

How cultural differences impacts information-seeking behavior and the recollection of information on websites

by

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CHAPTER ONE

Introduction

The development of the Internet has fundamentally changed how people look for information, communicate, and engage with one another. As websites become the central platform for knowledge sharing and service provision, optimizing their functionality and design is imperative for a seamless user experience. A crucial aspect of this optimization lies in understanding users' visual interactions with website content. It is known that culture significantly influences how people see and understand website content. Cultural backgrounds influence users' cognitive functions, attentional focus, and visual perception, resulting in various information-processing strategies. People from collectivist societies could tend to receive information more holistically and contextually, taking into account the implications and deeper meaning of the content.

People from individualistic cultures, meanwhile, could concentrate on details and distinctive features. According to Nisbet *et al.* (Nisbett and Masuda 2003) from psychological and cognitive science, westerners focus on specific objects, carefully analyzing their attributes and categorizing them to understand the underlying rules governing their behavior. In contrast, East Asians tend to have a broader perspective, focusing on the perceptual and conceptual fields. They notice relationships, changes, and group objects based on their family resemblance rather than rigid category membership.

Usually, in the graphical user interface design domain, there is a prevailing assumption that all end-users will perceive the presented information uniformly. However, this notion has come under scrutiny due to an expanding body of research in psychology and neuroscience, revealing that a person's cultural background plays a substantial role in shaping visual perception. Culture influences how individuals integrate contextual information into their perceptual processes. In order to produce user-centered designs that appeal to various audiences, website designers and developers must thoroughly understand various cultural influences.

Human-Computer Interaction (HCI) is a multidimensional space incorporating computer science, psychology, design, engineering, and social sciences elements. It includes both the study of humans and the study of systems. In this age of globalization with interactive computer-based technology, HCI research must study the cultural challenges that permeate and influence these technologies' design, development, evaluation, and usage. Some key factors to consider when developing technology are user needs, values, preferences, habits, and behavioral patterns, which cannot be understood apart from their cultural environment.

While in the past, the design viewpoint of "look and feel" was the primary concern in the field of HCI, presently social, political, and cultural issues are added to the aesthetic points of view, necessitating more complicated and in-depth study. The current field of HCI focuses on interaction design, experience design, societal concerns, usability, and many more (Hochheiser and Lazar 2007), (Garrett 2022), (Morville 2004), (Saffer 2009). User Experience (UX) is more than simply a system's usability and functioning; it also includes the social, emotional, and aesthetic components of the experience. To understand users' need, preferences, and behavior, HCI researchers design models and tools that includes cultural consideration while designing any system.

According to Hofstede *et al.* (Hofstede, Hofstede, and Minkov 2005), culture binds people through shared beliefs, values, and customs, making it challenging to quantify and evaluate its impact subtly. Culture is difficult to quantify and evaluate since its impacts on users are subtle, yet its effect and worth remain constant even in today's technologically dominated world. Despite the overwhelming nature of technological advancements, cultural factors continue to shape the purpose of technology consumption. Culture significantly impacts the outcomes of human-computer interactions and visual information processing (Salgado, Pereira, and Gasparini 2015). According to neurocognitive researchers, human visual perception varies due to culturally conditioned selective attention and memory processes (Gelfand, Chiu, and Hong 2015; Gutchess, Welsh, Boduroĝlu, and Park 2006). Boduroglu *et al.* (Boduroglu, Shah, and Nisbett 2009) used a visual change detection paradigm to investigate how people from different cultures allocate their attention while processing visual scenes . They found that East Asians are more sensitive to color changes than Americans. Other research demonstrates that different cultures' practices and beliefs support highly varied cognitive and attentional capacities while judging a target item. They showed that individuals who engage in Asian cultures better absorb contextual knowledge, while those who engage in North American cultures are better at ignoring it (Kitayama, Duffy, Kawamura, and Larsen 2003). Baughan *et al.* (Baughan, Oliveira, August, Yamashita, and Reinecke 2021) found that finding information on very complex websites takes three times as long for Japanese as it does for Americans . Dong *et al.* (Dong and Lee 2008) presented early evidence that differences in viewing patterns may affect website performance . For example, Chinese and Koreans scanned websites circularly in their investigation, but Americans browsed different screen sections sequentially.

Given these cultural distinctions in memory, attention, and visual preferences, this study aims to investigate whether these distinctions impact information-seeking behavior and recollection of information on websites. Specifically, the research investigates how users' choices concerning website visual complexity affect their search efficiency and which portions of a webpage they remember. Our research question is how users' (who view the content of a website) choice for different levels of website visual complexity affects their search efficiency and which portions of a page they remember. We determined this by measuring recall accuracy and searching time. The visual structure of the websites controls these two factors.

In order to investigate the variations in visual attention patterns, we conducted an online study involving 87 participants from Western cultures and 23 from Eastern cultures. During the study, participants were presented with website screenshots of varying complexities and instructed to perform specific information search tasks. Additionally, we examined whether they directed their attention to other sections of the websites while engaged in the primary search task. Our investigation explored whether these cross-cultural variances in attention patterns influence individuals' website navigation, specifically regarding their speed in locating information and their ability to recall it later. We assessed these factors based on the following:

- Participant's country of origin.
- The location of the information on the website, distinguishing between foreground and background elements.
- The complexity of the website, considering the density of text and images present on the site.

The investigation seeks to elucidate the influence of cultural backgrounds on visual attention patterns during website interactions. It will provide us with resources for a culturally-appropriate adaption for structured, tailored, and customized website design.

CHAPTER TWO

Related work

2.1 Culture and web design

Culture is a broad and complex concept to understand and define. It's a multi-variable, complex component frequently applied to describe human characteristics. Hundreds of definitions exist due to the high complexity of culture. Hofstede's definition is perhaps the most extensively used in the cross-cultural design domain. According to Professor Geert Hofstede, culture is a "collective programming of the mind which distinguishes the members of one human group from another" (Hofstede, Hofstede, and Minkov 2005). This collectiveness refers to how individuals of a cultural group share common meanings, patterns, norms, values, behavior, and practices (Sun 2012; Liu and Keung 2013). People from various cultural backgrounds perceive and respond in different ways. When these distinctions are examined closely, they go far beyond speaking and writing in other languages. Many factors reflect cultural differences, such as values, interactions, communication styles, visual preferences, and cognitive styles.

All these factors impact the design process of different user interfaces for different cultures (Plocher, Rau, Choong, and Guo 2021). According to Nordhoff *et al.* (Nordhoff, August, Oliveira, and Reinecke 2018) compared local and international websites across the 80,901 website designs from 44 nations, finding significant differences in design elements such as the website's colorfulness, visual complexity, the number of text areas, and the average saturation of colors. The design of websites with a global reach is more homogenized compared to local websites between countries. The study provides helpful pointers for website designers to customize their work for different countries, facilitating effective communication with various international consumers. Careful consideration of these factors is needed at the beginning of a design process and throughout the making of an application. Li *et al.* (Li, Karreman, and De Jong 2022) contributed to researching cultural differences in web design. They concluded that Western websites have more footers, but Chinese websites often have more items and active components. They found that links that open in new windows and the colors red and blue are frequently seen on Chinese websites. Western websites, on the other hand, frequently use white hues and highlighted links. Their findings also show that Western websites often show photographs of everyday life or the natural world, whereas Chinese websites frequently exhibit imagery connected to leaders or rituals.

2.2 Hofstede's cultural dimension model

Hofstede found six dimensions in his research that can be used to identify nations based on their cultures (Hofstede, Hofstede, and Minkov 2005).

(1) Power Distance: Power distance represents the extent of adherence to formal authority channels and is the degree to which the lesser powerful accept the prevailing distribution of power. High power distance cultures have members who are much more comfortable with centralized power than members of low power distance cultures.

(2) Individualism/Collectivism: In individualistic societies people take care only themselves. In collective societies people feel as though they belong to a strong group and always try to protect it. People with high individualism view self and immediate family as relatively more important than the collective.

(3) Masculinity/Femininity: In masculine cultures, winning is important and in feminine cultures, the welfare of disadvantaged members is taken care of.

(4) Uncertainty Avoidance: Uncertainty avoidance is the extent to which the members of a culture feel threatened by uncertain or unknown situations and try to avoid such situations.

(5) Long-term/Short-term Orientation: Long-term oriented societies attach more importance to the future. They foster pragmatic values oriented towards rewards, including persistence, saving and capacity for adaptation. For short term oriented societies, values promoted are related to the past and the present, including steadiness, and respect for tradition.

(6) Indulgence/Restraint: Indulgence stands for a society that allows relatively free gratification of basic and natural human drives related to enjoying life and having fun. Restraint stands for a society that suppresses gratification of needs and regulates it by means of strict social norms.

The initial research in the field of cultural sensitivity is mostly done by Marcus, Gould and Simon (Marcus and Gould 2000), (Simon 2000) and (Marcus 2003). A lot of studies found differences between websites from different countries. A lot of studies found the impact of culture on users' acceptance and ease of use. The study findings by Cerm´ak *et al.* (Čermák et al. 2020) establish a correlation between website elements and Hofstede's cultural dimensions. For example, headlines hold significance in countries with high individualism, low power distance, low uncertainty avoidance, and low indulgence. Newsletters, on the other hand, are associated with high indulgence and low long-term orientation, while search options are linked to high power distance cultures. As per research by (Radziszewska 2019), the dimensions of culture play a significant role in shaping the design of e-commerce websites. This influence stems from the fact that local cultures have a substantial impact on consumer online behavior and the desired quality attributes of Business-to-Consumer (B2C) websites. These studies have clearly shown that websites are culturally sensitive and used website elements differ according to the country of origin.

2.3 Effect of culture

Culture plays an important role in using and designing human-computer systems. It shapes human perception and cognition, perception of the environment and aesthetic preferences, self-esteem and information processing by offering sets of values, life expectations, and needs (Kastanakis and Voyer 2014). The way people from different cultures use website design elements also varies (Alexander, Thompson, and Murray 2017). People from different cultures shows differences in visual processing, understanding, and exploration of the picture content before selecting their picture passwords (Constantinides, Pietron, Belk, Fidas, Han, and Pitsillides 2020). Subjective usability ratings can vary depending on the participant samples' national culture (Sauer, Sonderegger, and Álvarez 2018). Here are more examples of how culture affects things.

2.3.1 Cultural effect on searching/attention mechanism:

Searching is an important component in web design to assist focused navigation on webpages. The efficiency of users' website searches can be influenced by their nationality (Baughan, Oliveira, August, Yamashita, and Reinecke 2021). The innate thinking processes generated by varied cultural backgrounds are believed to be the cause of variances in user search behavior during a website search. Many studies proved that Western culture shows object-focused attention mechanisms, whereas eastern Asian culture shows context-oriented attention during searching for information in websites (ref to be added). For GUI design, design-specific guidelines exist for designers to follow while designing a layout, component, and navigation among components. However, cross-cultural GUI design necessitates extra care because it must meet the needs and expectations of each user. Culturally sensitive design components have been demonstrated to attract and engage customers, as well as produce positive cognitive and sensory user experiences (Nielsen 2000).

2.3.2 Cultural effect on perspective taking:

The role of cultural influences has been extensively observed across a diverse range of academic disciplines, including business, science, psychology, and sociology. Notably, cultural differences exert a profound impact on individuals' self-perception and their perception of others (Kastanakis and Voyer 2014). Cultural norms also shape individuals' emphasis on social connections (Kastanakis and Balabanis 2012), as well as their decision-making processes and judgmental thinking (Matsumoto 2002). Findings from Wu *et al.* (Wu and Keysar 2007) highlight that collectivistic cultures, such as the Chinese, exhibit a tendency towards interdependence, directing individuals' attention towards others and consequently enhancing their perspective-taking abilities compared to individualistic cultures like that of Americans. While both cultural groups can discern between their own perspectives and those of others, the Chinese cultural context facilitates more effective utilization of this ability in interpreting the behaviors of others. The integration of culture's conditioning effects on perception and cognition in recent cross-cultural research presents an invaluable opportunity for researchers to gain deeper insights into cross-cultural consumer research and the development of marketing strategies across various domains.

2.3.3 Cultural effect on information processing:

Cultural factors profoundly shape individuals' cognitive processes, influencing how they perceive, interpret, and retain information. Eye movements and attentional focus have been reported to vary among cultures (Chua, Boland, and Nisbett 2005), showing the influence of cultural practices on the formation of perceptual habits. It also significantly impacts cognitive styles, decision-making processes, problem-solving methodologies, and team dynamics (Nisbett and Masuda 2003). Memory processes are significantly affected by culture as well. Culturally accepted knowledge, norms, and practices influence what people encode and remember (Conway, Wang, Hanyu, and Haque 2005). Additionally, cultural narrative traditions and storytelling practices impact memory structure and retention. Cross-cultural studies on memory have highlighted variations in memory recall accuracy and strategies, indicating the impact of cultural factors on this cognitive process.

2.3.4 Cultural effect on Q&A website:

Culture also has an impact on users' participation in Q&A websites (Oliveira, Andrade, and Reinecke 2016). Different people of diverse expertise have various reasons to contribute to these websites. Some people spend more time, and they have their reasoning and preference on what they want to work on (Nam, Ackerman, and Adamic 2009). In a study on Stackoverflow, which is a popular professional and enthusiast programmers website for Q&A, Schenk and Lungu (Schenk and Lungu 2013) found that people from North America and Europe made the most contributions. According to Hofstede's cultural dimension, people from the US and other western nations are more individualistic. According to Kayes *et al.* (Kayes, Kourtellis, Quercia, Iamnitchi, and Bonchi 2015), individualistic people provide more answers on the Q&A websites than in collectivists countries like China. Culture also affects digital image tagging behavior in the field of data mining. People from different cultures exhibit different tagging patterns. (Dong and Fu 2010) demonstrated that the first tag in a digital image was assigned to the main foreground object by European Americans. In contrast, the first tag was assigned to the general description in the image for Chinese people.

2.4 Fitts' Law

Fitts' Law (MacKenzie 1992) is a predictive model used to analyze pointing tasks in user interface design and human-computer interaction. It quantifies the time required to move between points, such as clicking a button. The formula indicates that the task becomes more challenging as the target becomes farther away and smaller. The movement time is linearly correlated with the Index of Difficulty (ID), which considers the distance to be covered and the target size. Fitts' Law helps inform the design of user interfaces, input devices, and pointing tasks to optimize usability and efficiency.

The formal equation for Fitts' Law is as follows:

$$\mathrm{MT} = a + b \cdot \mathrm{ID} = a + b \cdot \log_2 \left(\frac{2D}{W} + 1 \right) \tag{2.1}$$

Here MT represents the movement time required to reach the target. a and b are empirically derived coefficients. D represents the distance between the starting point and the center of the target. W represents the width of the target. The logarithmic term $\log_2 \left(\frac{2D}{W} + 1 \right)$ captures the task's difficulty, considering the distance to be covered and the target size. Using a logarithm, Fitts' Law accounts for the speed-accuracy trade-off in human movement, acknowledging that the task becomes more challenging as the distance increases or the target size decreases. The coefficients a and b are determined through empirical studies and depend on the specific setup and user population being analyzed. These coefficients allow for the calibration of the equation to match observed movement times in experimental data.

In examining whether an interface conforms to Fitts' Law, linear regression is performed over MT on the corresponding ID. The outcome of this regression analysis is an equation in the form of $\mathrm{MT} = a + b \cdot \mathrm{ID}$; where a is the intercept term, b is the slope of the regression line, and ID is the Index of Difficulty. The intercept term a represents the estimated time to navigate to a target when the Index of Difficulty (ID) is at its lowest level. $\mathrm{ID} = 0$ corresponds to the easiest task. A desirable outcome for intercept a is to be close to zero and if positive, ideally below 400 ms (Soukoreff and MacKenzie 2004). After the regression analysis, the coefficient of determination (R^2) is typically reported as a measure of the "goodness of fit". R^2 signifies the strength of the association between the ID and MT. R^2 indicates the extent to which the regression model explains the variability in the data, and higher R^2 values imply a better fit of the model.

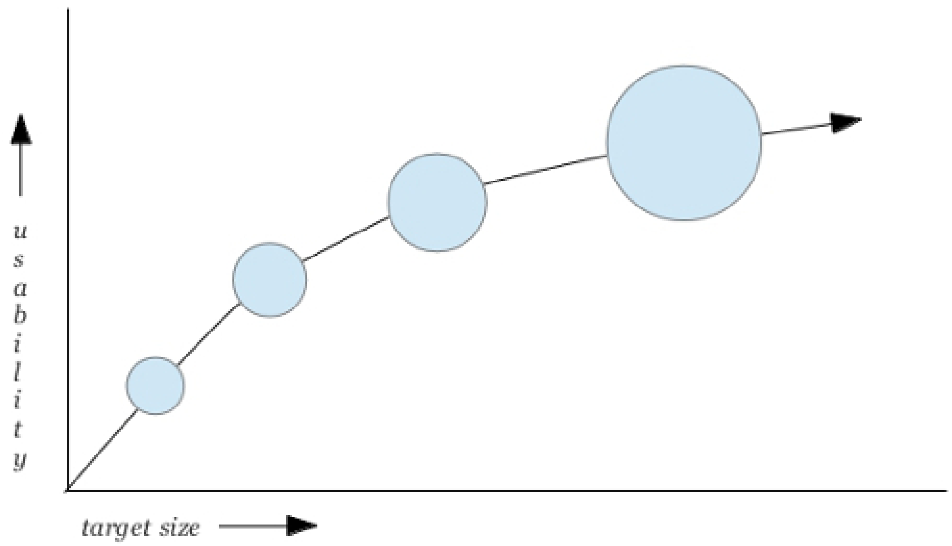

Figure 2.1. How fitts' law explains the relationship between usability and target size

Fitts' Law has been widely used in kinematics, human factors, and human-computer interaction in designing user interfaces, input devices, and pointing tasks, helping to optimize the arrangement and sizing of targets to improve usability and efficiency. It provides valuable guidelines for creating graphical user interfaces. Figure 2.1 shows that increasing the size of a button reduces the time required to click it and thus increases usability. As the button size gets larger, the effects are smaller; until the change eventually becomes marginal. Conversely, decreasing the button size leads to longer click times, especially if the button is small. Fitts' formula enables calculating the average time needed to click a button. Fitts' Law also suggests making frequently pressed buttons larger than less frequently pressed ones. Some key suggestions for fitts' law analysis are:

(1) Increasing target size enhances interaction efficiency. However, caution is necessary as larger targets may compromise overall usability and visual balance.

(2) Closer targets are acquired more swiftly. However, careful consideration is essential when elements are closely positioned to avoid inadvertent clicks on neighboring items.

(3) Fitts' Law facilitates the classification of input methods based on the physical effort they demand, enabling the identification of more efficient interaction techniques.

(4) Leveraging prime pixels, which are quicker to acquire, aligns with Fitts' Law recommendations to optimize target placement and improve overall user performance.

Our study integrated Fitts' Law analysis to address several key objectives. Given the various input devices used by our users, we aimed to determine the design that yielded faster and more accurate pointing actions. Additionally, we investigated the variation in click times concerning target size and assessed which design facilitated the selection of the most optimal interface for a given task. Applying Fitts' Law from a user experience standpoint offered valuable insights into potential usability issues and interface design bottlenecks. By comprehending the interplay between target size and distance, as explained by Fitts' Law, we aimed to enhance the efficiency and precision of pointing interactions within our study.

By considering all of these previous works, we presented two primary hypotheses. These hypotheses suggest differences in the speed at which participants locate information and the specific regions of a webpage they focus on while engaged in the search task. Our two hypotheses compares eastern and western countries based on two variables: search time and recall accuracy.

Hypothesis H1a: People from the East find information in the background of a website faster than people from the West, and vice versa for a website's foreground area.

Hypothesis H1b: How does the complexity of a website influence search time and moderate its association with the user's country.

Hypothesis H2: There is no association between participant's country, target area, website complexity and memory recall.

CHAPTER THREE

Methods

3.1 Participants:

Participants in this study were drawn from two sources. A subset of the participants were volunteers recruited by university campus announcements, online platforms, and community advertisements; while the remainder were Baylor University students who chose to participate in the study for additional credit. The study comprised 110 participants, representing a diverse sample for comprehensive analysis. There were 23 participants from Eastern cultural backgrounds, while the others were mainly from the United States, representing Western culture. Figure 3.1 shows the demographic breakdown of countries.

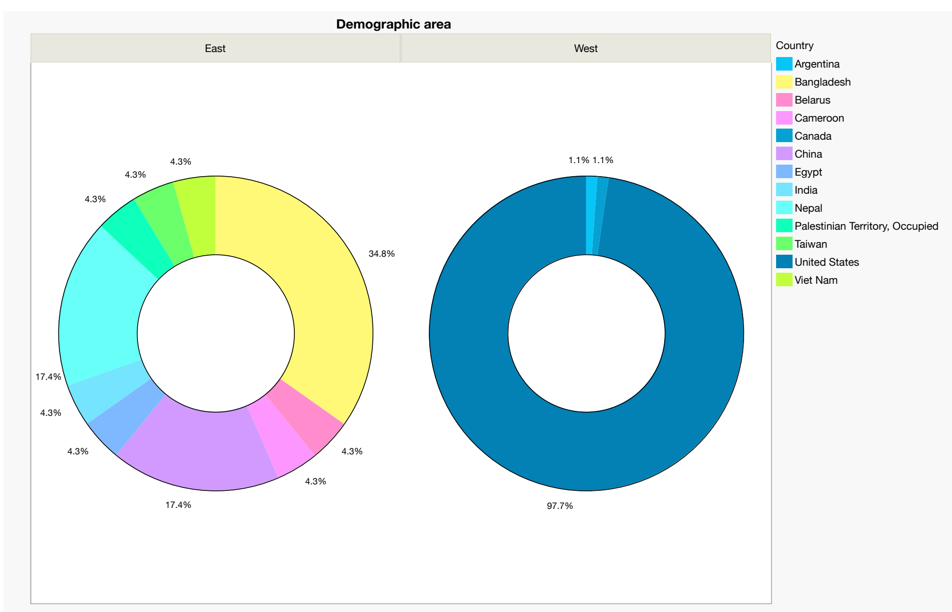

Figure 3.1. Pie chart of participants from different countries. From the eastern group, there were 10 countries, while from the western group, there were 3 countries.

Country	Eastern	Western
Number of participant	23	87

Gender		
female	10	31
male	13	55
non-binary	0	1

Input Device		
Mouse	43.48%	25.29%
Touch Screen	13.04%	11.49%
Touchpad	43.48%	63.22%

Age	Mean	23.953	20.13

Educational level		
Bachelor degree	30.43%	3.45%
Graduate degree	17.39%	1.15%
High school degree or equivalent (e.g., GED)	26.09%	39.08%
Postgraduate degree	8.70%	0.00%
Some college but no degree	17.39%	56.32%

Hours on computer daily	Mean	8.43478	6.6896

Figure 3.2. Overview of participant demographic

The recruiting strategy secured diverse participants from various cultural backgrounds, allowing the study to collect diverse viewpoints and user experiences. The participants' demographics were carefully chosen to investigate potential cross-cultural variations in user experience perceptions and preferences. In order to guarantee that participants were treated ethically, informed consent procedures were followed, and anonymity and confidentiality were maintained throughout the study. The Baylor IRB examined and approved the research methodology and participant recruiting procedure to verify that ethical principles were followed. The participant's mean age was 25.95 for the eastern group and 20.13 for the western group. The gender distribution was 51.7% men and 48.3% women, with one participant identifying as non-binary. On average, participants from eastern group spent 8.43 hours per day on a computer and participants from western group spent 6.69 hours per day on a computer, with the majority using a mouse as their primary input device, followed by a touchpad. Further details about the demographic breakdown are provided in Figure 3.2.

3.2 Materials and Procedure

From the top sites listed by Amazon Alexa, we chose twelve sites that had less exposure to the general public and covered a range of topics. We chose webpages for low, medium, and high complexity; four for each kind —according to the (Reinecke, Yeh, Miratrix, Mardiko, Zhao, Liu, and Gajos 2013) model of perceived visual complexity. In the model, Both Japanese and US participants displayed a shared inclination towards a comparable degree of visual intricacy (scoring 4.15 and 4.08 on a scale of 1 to 9, where 1 indicates minimal complexity and 9 signifies maximal complexity). Notably, individuals from both cultures exhibited a more adverse reaction to highly complex websites compared to simpler ones. However, it is noteworthy that Japanese participants demonstrated a slightly greater tolerance for highly intricate websites.

For task completion, we added simple tasks of clicking on a target for each webpage screenshot. For placing the "target" item, we set two areas on the webpage, foreground, and background. The "foreground" is the main item set against a wider scene backdrop, according to prior psychological studies (Masuda and Nisbett 2001). In order to adhere to this description, we defined the foreground area in our study as the main content area of a webpage. In contrast, the background area included all the other peripheral spaces on the page.

We started the experiment with a consent form, an overview of our experiment, and demographic questionnaires. We included one website screenshot and asked the user to complete a task. The task was to click on a target in the web page's foreground or background area. Targets were spread among the websites, half in the main content area and half in the background area.

Then we asked each user two questions. One question referred to a User Interface (UI) element in the main or background area. For example, "What was the button color you just clicked?" Each question had three options: "Correct", "Incorrect", and "I don't remember". The second question was on break tasks, which were included to flash participants' short-term memory. Break tasks are based on problem-solving tasks. The users were shown an image with a pattern with a missing piece. Then they were asked to select the pattern that best fits the missing piece.

After completing the break task, they were taken to a webpage screenshot image. In considering the inclusion of a "correct" or "incorrect" message for the break task segment, a question arises whether to include a "correct" or "incorrect" for its potential impact on participant encouragement. However, it was determined that this feature is outside the core objectives of the experiment. As a precautionary measure to mitigate any potential biases that could arise from such feedback, a decision was made to abstain from incorporating the "correct" or "incorrect" messages. The concern primarily revolves around the possibility of participant discouragement in the event of selecting an incorrect answer, which, in turn, might influence their subsequent responses.

3.3 Design

The experiment used a within-subject design. The dependent variables are search efficiency and information recall accuracy. Search efficiency was measured by the time between displaying the webpage screenshot and the user's click on the correct target. We accepted the click on the target as correct if they click within the five-pixel boundary of the target. Information recall accuracy is measured by the user's answers to questions about the UI element on the screenshot. It determines how much information a user can perceive while completing a task on that website.

Our independent variable is the participant's culture. We used the website's visual complexity and other demographic information as confounding variables. The study took 10 minutes on average.

CHAPTER FOUR

Result and Analysis

4.1 Data Anaysis

Regression/ANOVA model residuals with high degrees of skewness (symmetry) and kurtosis (peakedness) are undesirable and can invalidate these analyses. So, for each group, we analyzed the data to find how the data was distributed. (Hair 2009) and (Taris 2002) argued that data is normal if skewness is between -2 to +2 and kurtosis is between -7 to +7. Our data distribution showed that it is fairly normally distributed.

4.2 Hypothesis testing

4.2.1 ***H1a: People from the East find information in the background of a website faster than people from the West, and vice versa for a website's foreground area.***

To investigate our hypothesis, we employed a linear mixed-effects model. The model incorporated log-transformed search time as the dependent variable and participant name as a random variable, while country (west-east), target area (foreground-background), age, and input device served as the independent variables. Moreover, we analyzed the interaction effect between the country and the target area. The findings do not provide support for our hypothesis. We did not identify a statistically significant interaction effect between the target area and country. Overall, all participants exhibited slightly quicker search times for information located in the main area of a page compared to the periphery, as explained in figure 4.1 and corroborated by the model results, wherein "target area" emerged as a significant factor.

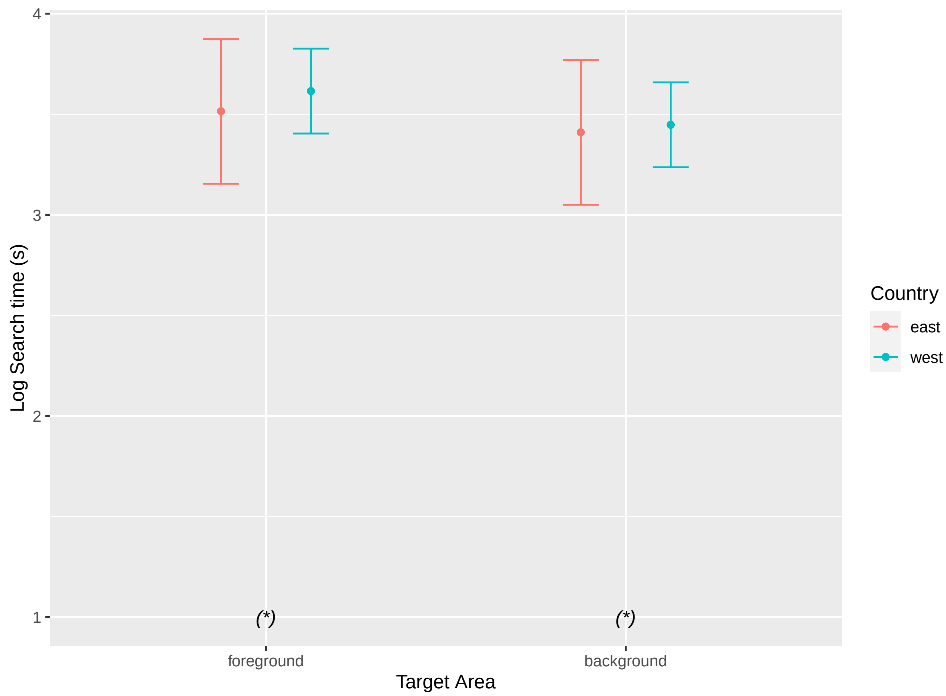

Figure 4.1. The marginal means of log-transformed search time versus task type. Bars show confidence intervals at 95%, and asterisks indicate significant differences with p<.001.

Figure 4.1 shows the marginal means of log-transformed search time versus task type show no interaction effect between country and target area (disconfirming H.1a). In the previous research study by (Baughan, Oliveira, August, Yamashita, and Reinecke 2021), US Americans were significantly faster than Japanese participants when finding information in the periphery. But per our analysis, for both foreground and background target areas, people from both the cultural group shows similar search time.

The model showed how the country affects search time. There was no significant difference in search time between the two groups of participants. Also, the difference was not significant for websites where the target was in the main content area ($t = -0.18, p = 0.99$, and Cohen's $d = 0.10$) and where the target was in the periphery ($t = -0.49, p = 0.96$, and Cohen's $d = 0.04$). The p value means the

Table 4.1. Results of ANOVAs performed on two linear regression models to test the interaction effect of country and target area on search time (H.1a)

	F-statistics	p-value
country	0.2454	0.6211
target area	0.8200	0.3654
age	0.0.2517	0.6170
input device	4.1214	< 0.05
country x target area	0.2364	0.6269

probability, for a given statistical model that, when the null hypothesis is true, the statistical summary would be equal to or more extreme than the actual observed results (Greenland, Senn, Rothman, Carlin, Poole, Goodman, and Altman 2016). Cohen's d is a standardized effect size for measuring the difference between two group means (Cohen 2013). The model also showed an interaction effect of country and target area on search time. An interaction effect occurs when the effect of one variable depends on the value of another variable (Jaccard and Turrisi 2003). An F statistic is used to find out if the means between two populations are significantly different(Wright 1965). Table 4.1 shows the results of ANOVAs performed on two linear regression models to test it. It shows no interaction effect between country and target area on search time, disconfirming H.1a.

4.2.2 ***H1b: How does the complexity of a website influence search time and moderate its association with the user's country.***

We added an interaction effect between country and website complexity to the model used to test H.1a. We tested how website complexity correlates with search time and whether website complexity moderates the relationship between search time and country. The results confirm H.1b As shown in figure 4.2, there is a positive correlation between website complexity and search time: The more complex a webpage, the more time participants took to find information. The graph also showed that when the target area is in the foreground, participants from eastern countries

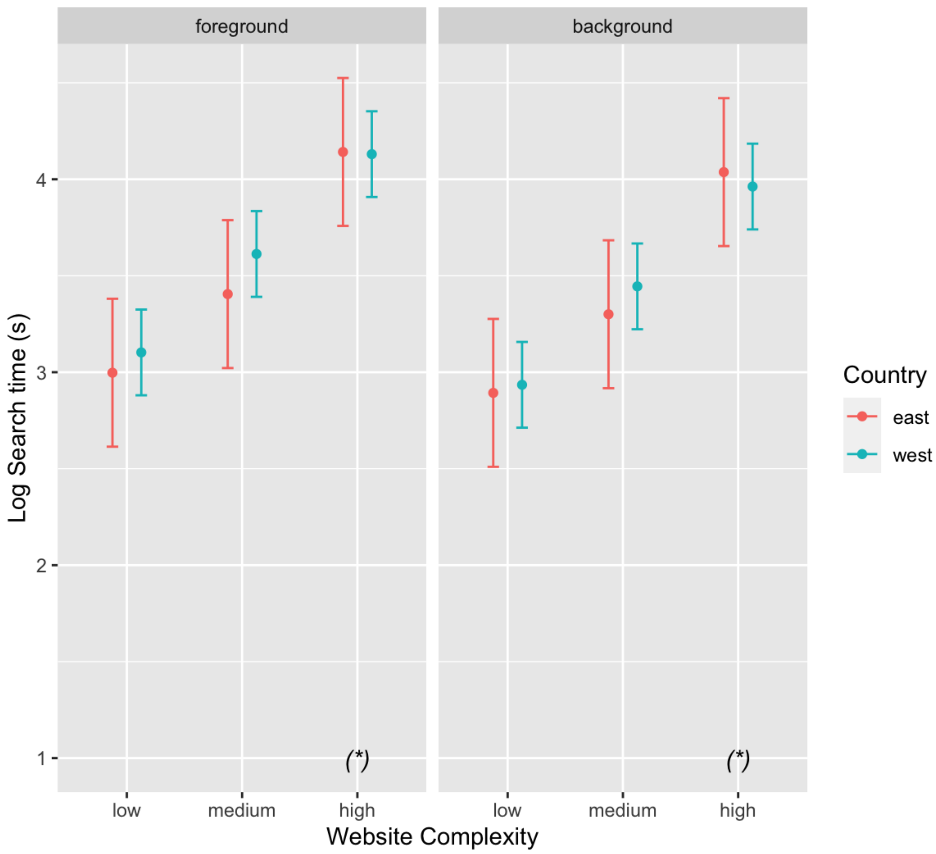

Figure 4.2. Search time increases with increasing website complexity. Bars show confidence intervals at 95%, and asterisks indicate significant differences with p<0.001.

required less time for the complexity low and medium, while for high-complexity websites, participants from both groups required almost the same time. When the target area was in the background, participants from western countries required less time for the complexity low and medium, while for high-complexity websites, participants from eastern countries required more time. This result support prior analysis from (Baughan, Oliveira, August, Yamashita, and Reinecke 2021) and (Baughan, August, Yamashita, and Reinecke 2020). Table 4.2 shows the results with a significant interaction effect between country and website complexity on search time, confirming H.1b.

Table 4.2. Results of ANOVAs performed on two linear regression models to test the interaction effect of country and website complexity on search time (H.1b)

	F-statistics	p-value
country	0.1291	< 0.001
target area	5.5149	0.72011
age	0.2491	0.61877
website complexity	117.8517	< 0.001
input device	4.1169	< 0.05
country x target area	0.2991	0.58452
country x website complexity	1.1941	0.30332

4.2.3 ***H2: There is no association between participant's country, target area, website complexity and memory recall.***

We performed a chi-square test to investigate the relationship between two categorical variables; country and right/wrong answer. Our null hypothesis was "There is association between country and memory recall variable." Our dependent variable was the probability of correctly answering a memory recall question.

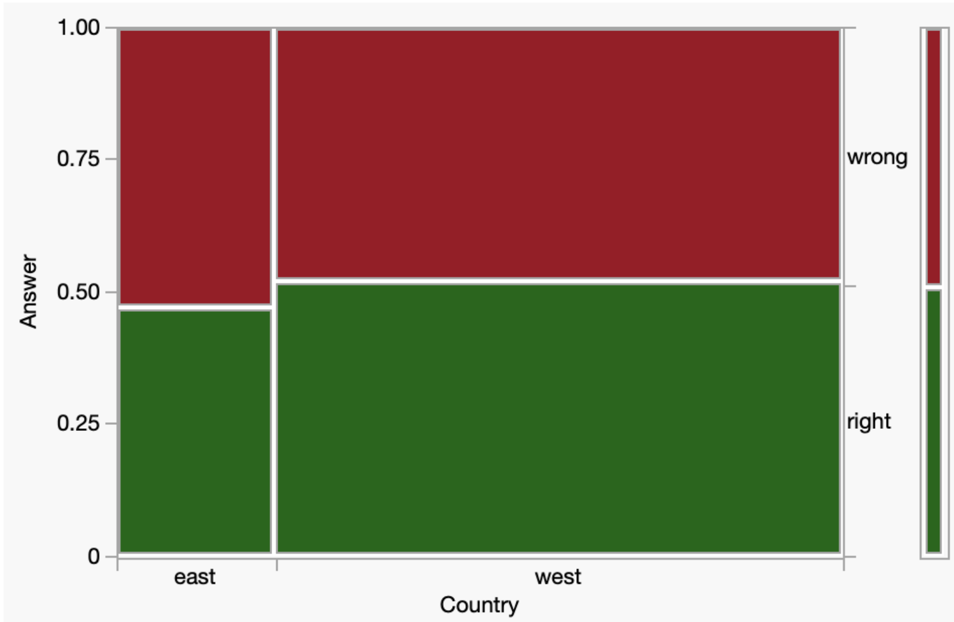

Figure 4.3. Answer for all questions vs country group shows no significant association between these two variables

Figure 4.3 shows a mosaic plot for our analysis. The mosaic plots shows us at once, the marginal distribution of country and answer type. It showed that the majority of participants from the East had answered incorrectly. In the case of participants from the West, there was an equal correlation between right and wrong answers. From the table in 4.3, we can see that the percentage of correctly answering and wrongly answering were mostly the same. Fisher's exact test (Fisher 1935) shows that the probability of answering "wrong" is greater for east than west. The Pearson and likelihood ratio (Rao 2002) tells us that there is no statistical significance between the two variables. As from table in 4.4 the p-value > 0.05, we can reject our null hypothesis and conclude that there is no association between these two factors.

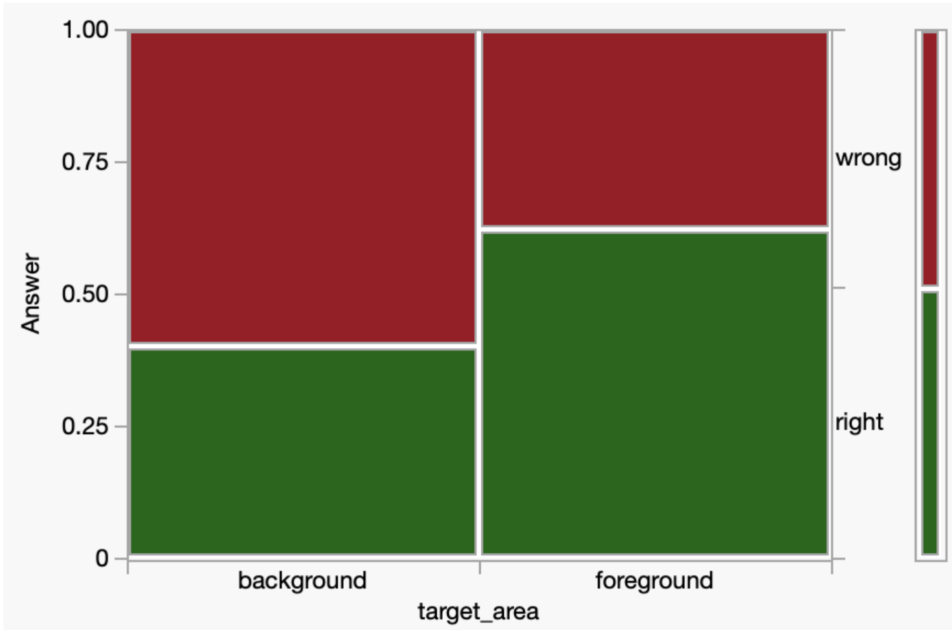

Figure 4.4. Answer for all questions vs target area shows significant association between these two variables

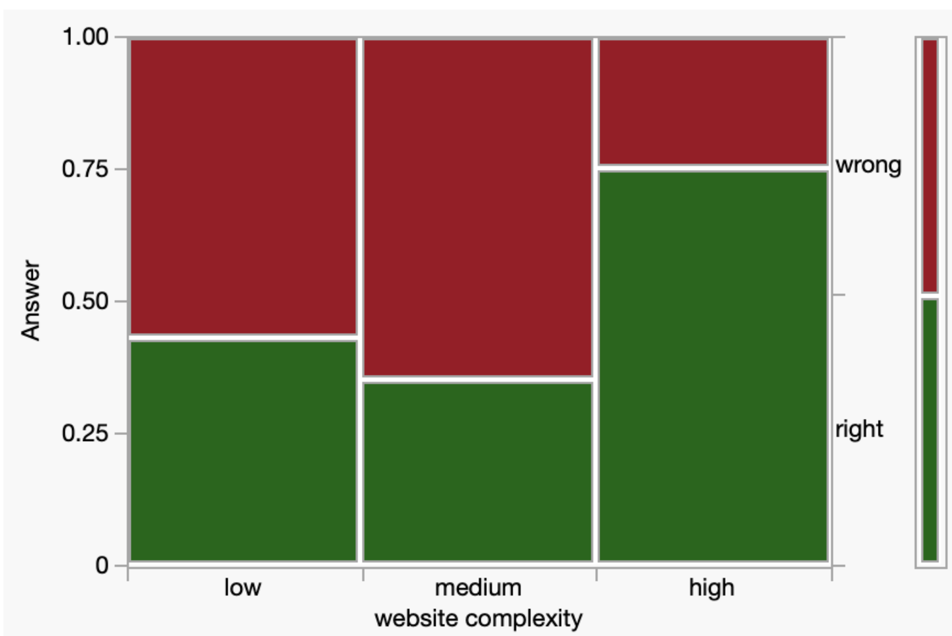

Figure 4.5. Answer for all questions vs. website complexity type shows significant association between these two variables

Figure 4.4 and 4.5 shows how probability of correct answers has been affected by target area and website complexity. A Chi-square test shows significance differences between these variables . When the target was in “foreground”, participants tend to correctly answer those. On the other hand, when website complexity was high, participants tend to answer correctly.

Table 4.3. Contingency table

Country	Right	Wrong
East	47.22%	52.78%
West	52.13%	47.87%

Table 4.4. Chi-Square test

Test	ChiSquare	Prob>ChiSq
Pearson	2.172	0.1406

4.3 Fitts' Regression

Figure 4.6 shows Fitts's law analysis that visualizes the impact of the Index of Difficulty on Movement Time. Linear regression performed on the whole data set shows that movement time varied independently of the difficulty index. The result is a regression equation in the form of Equation 2.1. This serves as a prediction of the time it takes to move when clicking on a target. For all of the input device type a value of $R^2 = 0.0137$ indicates that the regression line explains 1.37% of the variation, which is very small. The R-value is 0.117, which is < 0.5. R^2 value for touchpad is 0.0087 and for mouse is 0.021, which shows little variability. With an r-square coefficient close to 0, this data fails to corroborate Fitts' law totally.

The observed results can be attributed to several factors influencing the study's movement time. The time to read the question and locate the target impacts the

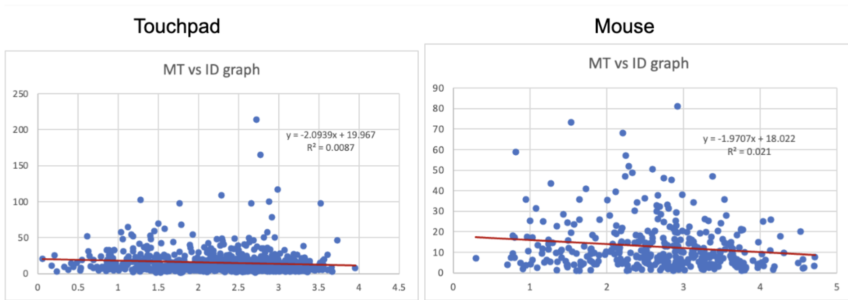

Figure 4.6. Movement time as a function of task difficulty when user input data was "mouse" and "touchpad"

overall movement time. Since the questions presented varied in difficulty levels, participants likely required varying durations to comprehend and process each question before initiating their search for the target. Furthermore, the targets were positioned in different areas throughout the interface, which likely affected the time to locate and accurately click on them. These variations in question difficulty and target placement contributed to the differences in movement time across the study.

According to Sambrooks and Wilkinson (Sambrooks and Wilkinson 2013), gestural interaction performed much worse than touch and mouse interaction. During the analysis process, we decided to omit data points where the user's input device was recorded as a "touchscreen." This exclusion was because, in these cases, users directly clicked on the target without interacting with any other part of the screen, resulting in a single data point with only an x-y coordinate. As our recommended input device for the study was the "mouse", we chose not to include the "touchscreen" data in the analysis to avoid potential anomalies in our results. This decision was made to ensure the consistency and reliability of the data, as the "touchscreen" data did not align with the primary focus on mouse interaction in our study.

4.4 Cultural dimension analysis

To compare the culturally influenced values and sentiments of similar respondents from these countries, we utilized the VSM (Values Survey module) by (Hofstede and Minkov 2013). In our study, we focused on two countries, Bangladesh and the USA for two reasons. Firstly, there was not enough data for all of the countries that our participants were from. Secondly, because these two country represented the majority of participants in our two groups. The comparison scores of six dimensions for both countries are presented in Figure 4.7. The graph indicates that our findings from the analysis section align with the Hofstede model for the selected countries. This supports the validity and applicability of the Hofstede model in our study's context.

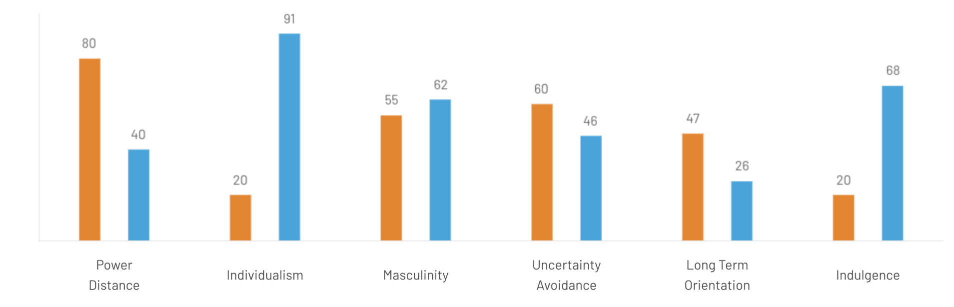

Figure 4.7. This bar graph shows the score of six cultural dimension of Bangladesh and USA, from where the number of participants were most from two group. The color "Orange" represents the country "Bangladesh" and the color "Blue" represents the country "U.S."

- **Power Distance:** Bangladesh obtains a high score of 80 on this dimension, indicating a strong acceptance of hierarchical order, where individuals are content with their designated places and do not require additional justification. Conversely, the USA receives a score of 40, indicating a comparatively lower level of equality, with individuals treating themselves more unequally.

- **Individualism:** Bangladesh is recognized as a collectivistic society, attaining a score of 20. Conversely, the United States represents a culture that embraces equal rights in all facets of American society and government, making it one of the most individualistic cultures globally, scoring 91 in this dimension.
- **Masculinity:** Bangladesh obtains a score of 55 on this dimension, classifying it as a Masculine society. In such societies, the focus lies on "living to work," emphasizing competition, achievement, and success, with the winner or top achiever being esteemed. On the other hand, a lower score on this dimension (Feminine) signifies that the prevailing values in society revolve around caring for others and improving the overall quality of life. Comparatively, Bangladesh scores slightly lower than the USA on this dimension.
- **Uncertainty Avoidance:** Bangladesh scores 60 on the Uncertainty Avoidance dimension, indicating a strong inclination toward maintaining rigid codes of belief and behavior. In such cultures, there is little tolerance for unconventional ideas or behaviors. People in high Uncertainty Avoidance societies have a strong emotional need for rules, even if they may not always be practical. Time is highly valued as money, and there is an inner urge to stay busy and work diligently. Precision and punctuality are the norm, while innovation might encounter resistance, as security plays a crucial role in individual motivation. On the other hand, the score of 46 for Americans suggests a relatively lower level of Uncertainty Avoidance, implying a greater openness to diverse ideas and opinions. Freedom of expression is generally respected in their culture, allowing for a more accepting attitude toward various perspectives.
- **Long-Term Orientation:** This dimension pertains to how societies balance preserving their links with the past while addressing present and future challenges. Bangladesh receives an intermediate score of 47 on this dimension,

suggesting it does not strongly prefer either direction. In other words, the country maintains a relatively balanced approach to managing its historical connections and adapting to current and future circumstances. In contrast, the United States scores lower at 26, indicating a tendency to analyze new information to verify its validity critically. This suggests that the United States is more inclined toward embracing new ideas and concepts while evaluating their credibility and relevance.

- **Indulgence:** This dimension measures the level to which individuals regulate their desires and impulses, influenced by their upbringing. Bangladesh obtains a very low Indulgence score of 20, categorizing it as a Restrained country. In societies with low Indulgence scores, there tends to be a tendency towards cynicism and pessimism. On the other hand, the United States scores as an Indulgent society (68). This indicates a culture that values a "work hard, play hard" mentality, allowing for more indulgence in satisfying desires and enjoying life's pleasures.

CHAPTER FIVE

Discussion

Researchers in psychology and neuroscience have questioned the widely accepted belief that visual attention patterns are uniform across individuals. Their studies have demonstrated that culture plays a significant role in influencing where people direct their attention, subsequently impacting the aspects of an image or scene they tend to remember. This study investigates whether these divergent visual attention patterns could influence search efficiency and recall when navigating websites.

Our analysis did not support our initial hypothesis, indicating that individuals from both the Eastern and Western generally exhibit similar search time on information retrieval in both target areas (background and foreground) of a webpage. This result did not align with a previous study conducted by (Dong and Lee 2008), which observed that East Asians tend to scan websites circularly, leading to quicker search times. (Wang, Masuda, Ito, and Rashid 2012) also found that East Asians outperformed Westerners in finding information on longer mock websites without textual information. However, both groups showed similar speeds on shorter websites, which our study corroborated. However, the study by (Baughan, Oliveira, August, Yamashita, and Reinecke 2021) reported that US Americans were faster than the Japanese at finding information in both the main content area and periphery, contrasting our results. Also, we did not find a correlation between the performance difference between East and West participants and website complexity.

Regarding our second hypothesis, which drew from prior research by (Chua, Boland, and Nisbett 2005) and (Masuda and Nisbett 2001) suggesting that East Asians focus more on contextual information, our analysis only partially confirmed it. While we found similarities in how both cultural groups recalled information, thus not entirely supporting the context sensitivity theory, the recall accuracy was influenced by the target area and website complexity. Participants were more likely to answer correctly when the target was in the foreground area, and the website's complexity was higher. This indicates that participants devoted more time to correctly identifying the target in high-complexity websites, leading to a higher proportion of correct answers.

To explain the disparities between these findings, we considered two potential factors. Firstly, unlike the previous study, which focused solely on Japanese participants, our study included participants from various countries within both cultural groups. This diverse participant pool could have influenced our research outcomes. Secondly, language played a role; our experiment utilized English-only websites, whereas the study by (Baughan, Oliveira, August, Yamashita, and Reinecke 2021) involved translated websites, and reading speed in different languages might have impacted the results.

CHAPTER SIX

Limitations and future work

In future research, several areas can be explored to enhance our understanding of the topic:

Effect of Language: Investigate how language impacts visual attention and search efficiency. Comparing participants' performance on websites in different languages could shed light on the influence of linguistic factors on information retrieval.

Relation between Design and Task Speed: Explore the relationship between website design elements (e.g., layout, color scheme, font size) and participants' speed in completing search tasks. Understanding how design choices affect user behavior can inform the development of more user-friendly websites.

Large set of Websites: Increase the number of websites used in the study to obtain a more comprehensive analysis. A diverse set of websites from various domains and styles would provide a broader perspective on the generalizability of the findings.

Eye Tracking Study: Conduct an eye-tracking study alongside the search task to gain deeper insights into participants' visual attention patterns. Eye tracking data can reveal precise gaze points and fixation durations, enhancing our understanding of how attention is distributed across web elements.

By incorporating these aspects into future investigations, we can gain valuable insights into the complexities of visual attention, search behavior, and website design, contributing to improving user experiences and web usability.

CHAPTER SEVEN

Conclusion

To address the question of cultural differences in website design, we conducted an online study involving participants from eastern and western countries. There were given the task of completing the task of finding a target, with the information presented either in the foreground or background of a webpage. Our research yielded interesting insights, revealing distinct approaches among people from different cultures when interacting with websites. By shedding light on these differences, our study aims to contribute to the field of website design, emphasizing the importance of considering varying cultural preferences to enhance user experiences and optimize search efficiency across different cultural backgrounds. For example, focusing on clean layouts, intuitive navigation, and user-friendly interfaces can enhance the user experience for both Western and Eastern audiences. Since search time did not differ significantly, designers should emphasize on efficient information retrieval techniques that cater to users' cognitive processes, such as implement clear labeling, structured menus, and easy-to-use search functions to facilitate quick access to desired information. While search time for information on webpages appears consistent across Western and Eastern cultures, a holistic approach to website design that emphasizes clarity, efficiency, adaptability, and cultural inclusivity remains crucial for creating a positive user experience across diverse audiences.

BIBLIOGRAPHY

Alexander, R., N. Thompson, and D. Murray (2017). Towards cultural translation of websites: a large-scale study of australian, chinese, and saudi arabian design preferences. *Behaviour & Information Technology 36*(4), 351–363.

Baughan, A., T. August, N. Yamashita, and K. Reinecke (2020). Keep it simple: how visual complexity and preferences impact search efficiency on websites. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*, pp. 1–10.

Baughan, A., N. Oliveira, T. August, N. Yamashita, and K. Reinecke (2021). Do cross-cultural differences in visual attention patterns affect search efficiency on websites? In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*, pp. 1–12.

Boduroglu, A., P. Shah, and R. E. Nisbett (2009). Cultural differences in allocation of attention in visual information processing. *Journal of Cross-Cultural Psychology 40*(3), 349–360.

Čermák, R. et al. (2020). Culturally sensitive website elements and features: A cross-national comparison of websites from selected countries. *Acta Informatica Pragensia 9*(2), 132–153.

Chua, H. F., J. E. Boland, and R. E. Nisbett (2005). Cultural variation in eye movements during scene perception. *Proceedings of the national academy of sciences 102*(35), 12629–12633.

Cohen, J. (2013). *Statistical power analysis for the behavioral sciences*. Academic press.

Constantinides, A., A. M. Pietron, M. Belk, C. Fidas, T. Han, and A. Pitsillides (2020). A cross-cultural perspective for personalizing picture passwords. In *Proceedings of the 28th ACM Conference on User Modeling, Adaptation and Personalization*, pp. 43–52.

Conway, M. A., Q. Wang, K. Hanyu, and S. Haque (2005). A cross-cultural investigation of autobiographical memory: On the universality and cultural variation of the reminiscence bump. *Journal of cross-cultural psychology 36*(6), 739–749.

Dong, W. and W.-T. Fu (2010). Cultural difference in image tagging. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*, pp. 981–984.

Dong, Y. and K.-P. Lee (2008). A cross-cultural comparative study of users' perceptions of a webpage: With a focus on the cognitive styles of chinese, koreans and americans. *International Journal of Design 2*(2).

Fisher, R. A. (1935). The logic of inductive inference. *Journal of the royal statistical society 98*(1), 39–82.

Garrett, J. J. (2022). *The elements of user experience.* .

Gelfand, M. J., C.-y. Chiu, and Y.-y. Hong (2015). *Handbook of Advances in Culture and Psychology, Volume 5.* Oxford University Press.

Greenland, S., S. J. Senn, K. J. Rothman, J. B. Carlin, C. Poole, S. N. Goodman, and D. G. Altman (2016). Statistical tests, p values, confidence intervals, and power: a guide to misinterpretations. *European journal of epidemiology 31*, 337–350.

Gutchess, A. H., R. C. Welsh, A. Boduroĝlu, and D. C. Park (2006). Cultural differences in neural function associated with object processing. *Cognitive, Affective, & Behavioral Neuroscience 6*(2), 102–109.

Hair, J. F. (2009). Multivariate data analysis.

Hochheiser, H. and J. Lazar (2007). Hci and societal issues: A framework for engagement. *International Journal of Human [# x02013] Computer Interaction 23*(3), 339–374.

Hofstede, G., G. J. Hofstede, and M. Minkov (2005). *Cultures and organizations: Software of the mind*, Volume 2. Mcgraw-hill New York.

Hofstede, G. and M. Minkov (2013). Vsm 2013. *Values survey module*.

Jaccard, J. and R. Turrisi (2003). *Interaction effects in multiple regression.* Number 72. Sage.

Kastanakis, M. N. and G. Balabanis (2012). Between the mass and the class: Antecedents of the "bandwagon" luxury consumption behavior. *Journal of business research 65*(10), 1399–1407.

Kastanakis, M. N. and B. G. Voyer (2014). The effect of culture on perception and cognition: A conceptual framework. *Journal of Business Research 67*(4), 425–433.

Kayes, I., N. Kourtellis, D. Quercia, A. Iamnitchi, and F. Bonchi (2015). Cultures in community question answering. In *Proceedings of the 26th ACM Conference on Hypertext & Social Media*, pp. 175–184.

Kitayama, S., S. Duffy, T. Kawamura, and J. T. Larsen (2003). Perceiving an object and its context in different cultures: A cultural look at new look. *Psychological science 14*(3), 201–206.

Li, Y., J. Karreman, and M. De Jong (2022). Cultural differences in web design on chinese and western websites: A literature review. In *2022 IEEE International Professional Communication Conference (ProComm)*, pp. 105–111. IEEE.

Liu, P. and C. Keung (2013). Defining cross-culture theoretical framework of user interface. In *International Conference on Cross-Cultural Design*, pp. 235–242. Springer.

MacKenzie, I. S. (1992). Fitts' law as a research and design tool in human-computer interaction. *Human-computer interaction 7*(1), 91–139.

Marcus, A. (2003). Are you cultured. *New Architect 8*(3), 28–31.

Marcus, A. and E. W. Gould (2000). Crosscurrents: cultural dimensions and global web user-interface design. *interactions 7*(4), 32–46.

Masuda, T. and R. E. Nisbett (2001). Attending holistically versus analytically: comparing the context sensitivity of japanese and americans. *Journal of personality and social psychology 81*(5), 922.

Matsumoto, D. (2002). Methodological requirements to test a possible in-group advantage in judging emotions across cultures: comment on elfenbein and ambady (2002) and evidence.

Morville, P. (2004). User experience design. *Semantic Studios 21*.

Nam, K. K., M. S. Ackerman, and L. A. Adamic (2009). Questions in, knowledge in? a study of naver's question answering community. In *Proceedings of the SIGCHI conference on human factors in computing systems*, pp. 779–788.

Nielsen, J. (2000). Designing web usability.

Nisbett, R. E. and T. Masuda (2003). Culture and point of view. *Proceedings of the National Academy of Sciences 100*(19), 11163–11170.

Nordhoff, M., T. August, N. A. Oliveira, and K. Reinecke (2018). A case for design localization: Diversity of website aesthetics in 44 countries. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems*, pp. 1–12.

Oliveira, N., N. Andrade, and K. Reinecke (2016). Participation differences in q&a sites across countries: opportunities for cultural adaptation. In *Proceedings of the 9th Nordic Conference on Human-Computer Interaction*, pp. 1–10.

Plocher, T., P.-L. P. Rau, Y.-Y. Choong, and Z. Guo (2021). Cross-cultural design. *Handbook of human factors and ergonomics*, 252–279.

Radziszewska, A. (2019). Cultural determinants of e-commerce website quality assessment. *Economic and Social Development: Book of Proceedings*, 17–25.

Rao, C. (2002). Karl pearson chi-square test the dawn of statistical inference. *Goodness-of-fit tests and model validity*, 9–24.

Reinecke, K., T. Yeh, L. Miratrix, R. Mardiko, Y. Zhao, J. Liu, and K. Z. Gajos (2013). Predicting users' first impressions of website aesthetics with a quantification of perceived visual complexity and colorfulness. In *Proceedings of the SIGCHI conference on human factors in computing systems*, pp. 2049–2058.

Saffer, D. (2009). The disciplines of user experience. *Designing for interaction.*

Salgado, L., R. Pereira, and I. Gasparini (2015). Cultural issues in hci: challenges and opportunities. In *International Conference on Human-Computer Interaction*, pp. 60–70. Springer.

Sambrooks, L. and B. Wilkinson (2013). Comparison of gestural, touch, and mouse interaction with fitts' law. In *Proceedings of the 25th Australian Computer-Human Interaction Conference: Augmentation, Application, Innovation, Collaboration*, pp. 119–122.

Sauer, J., A. Sonderegger, and M. A. H. Álvarez (2018). The influence of cultural background of test participants and test facilitators in online product evaluation. *International Journal of Human-Computer Studies 111*, 92–100.

Schenk, D. and M. Lungu (2013). Geo-locating the knowledge transfer in stackoverflow. In *Proceedings of the 2013 international workshop on social software engineering*, pp. 21–24.

Simon, S. J. (2000). The impact of culture and gender on web sites: an empirical study. *ACM SIGMIS Database: The Database for Advances in Information Systems 32*(1), 18–37.

Soukoreff, R. W. and I. S. MacKenzie (2004). Towards a standard for pointing device evaluation, perspectives on 27 years of fitts' law research in hci. *International journal of human-computer studies 61*(6), 751–789.

Sun, H. (2012). *Cross-cultural technology design: Creating culture-sensitive technology for local users.* OUP USA.

Taris, T. (2002). Bm byrne, structural equation modeling with amos: Basic concepts, applications, and programming mahwah nj: Lawrence erlbaum, 2001 0-8058-3322-6. *European Journal of Work and Organizational Psychology 11*, 243–246.

Wang, H., T. Masuda, K. Ito, and M. Rashid (2012). How much information? east asian and north american cultural products and information search performance. *Personality and Social Psychology Bulletin 38*(12), 1539–1551.

Wright, S. (1965). The interpretation of population structure by f-statistics with special regard to systems of mating. *Evolution*, 395–420.

Wu, S. and B. Keysar (2007). The effect of culture on perspective taking. *Psychological science 18*(7), 600–606.